Manual

Setup Planning

HLS-RWP 8.1

Version 1.0.23049

Last changed on: 01.09.2020

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1 Übersicht Rüstwechselplanung

Purpose

The function package provides the tools needed to specifically handle setup times resulting, for example, from tool, material or color changes.

Implementation considerations

You use the function package if you have setup-intensive production and want to reduce setup times during detailed scheduling.

Integration

By using setup change times, HYDRA shop floor scheduling can ascertain dynamic, i.e. occupancy-dependent setup times. Setup change times can be defined for tool, material, color or article changes.

Features

- Creation and input of a setup change matrix
- The ability to determine and plan sequence-dependent times for setup at workplaces / machines
- Differentiation by change of tool, color and product based on dynamic setup times that are defined in the setup change matrix.
- Automatic elimination of static setup times that were transferred from ERP if the same tools, colors and products were detected in operations scheduled to run one after the next
- Calculation of totals for setup and dismantling times (teardown)

2 Setup Change Matrix

Overview

Menu	Master data → Production control → Setup change matrix
Transaction code	setmx
Function authorization	mdsetmx

Purpose

You use this application to configure setup change times.

Integration

Using setup change times, HYDRA Shop Floor Scheduling can calculate dynamic, i.e. assignment-dependent setup times. You can define setup change times for tools, material, colors or articles.

Requirements

Depending on the area or scope of application, you must have entered and maintained data in the *tool, color, material* and/or *article* fields of the operation.

Selection criteria

The application provides the following selection criterion:

Type

You can use the combo box to restrict setup change times to

- Tool
- Color
- Material
- Article

Additionally, you can select specific user fields of the order header and operation. But you have to assign the user fields to the user field key SYSTEM when you configure the system. The system supports alphanumeric and integer user fields. Please note that unassigned integer user fields have the value 0. Further information on user fields can be found below.

Field descriptions

Type

Defines the type of setup change time. This might be: Tool change, material change or color change.

Group

Setup times apply specifically to a capacity group. This field can be left blank, in which case the dynamic setup time is the same across all capacity groups.

Workplace

If the setup times should only apply to one particular workplace within a capacity group, this must be specified here. In addition, the group the workplace is assigned to must be correctly specified. This field can be left blank, in which case the dynamic setup time is the same across all capacity groups.

From

Depending on the type selected, the corresponding value is defined here that should be used to determine the dynamic setup time. In this case, "from" means that the value is defined in the preceding operation.

To

Depending on the type selected, the corresponding value is defined here that should be used to determine the dynamic setup time. In this case, "to" means that the value is defined in the subsequent operation.

Dynamic setup time

Setup time that must be specified additionally because of the setup change. This value is stored in the field "Setup time addition" of the operation. If the value is negative, it is deducted from the static setup time (up to a max. of 0). When the operation is saved, the negative value is transferred and saved in the field "setup time addition"; the static time is saved unchanged.

Ignore static setup time

If this option is set and the "from/ to" criteria are met, the static setup time (preparation) is ignored.

Doing so - unlike for dynamic setup times - you can define that the "static setup time" transferred from the ERP system and defined at the operation should only be used if *no* change (e.g. color change) is made.

Processing notes

The check is run against the *tool, color, material* and/or *article* fields at the operation and the configured user fields of the order header or operation.

Case sensitivity

Value comparisons are generally case sensitive. Therefore, make sure to use correct spelling when configuring the setup change matrix and in the pool of operations data.

Example:

Type	From	To	Dyn. setup time
TOOL	W**z**1	WZ2	3:00
TOOL	WZ2	WZ1	5:00

OP 1		OP 2	
- TOOL:	WZ1	- TOOL:	WZ2
- COLOR:	YELLOW	- COLOR:	BLUE

If OP 1 (tool WZ1) is planned after OP 2 (tool WZ2), then it is given 5:00 hours of dynamic (additional) setup time. If, on the other hand, they are planned in opposite order (OP 2 after OP 1), then no dynamic setup time is used, because "W**z**1“ (not "WZ1“) is the first entry in the setup change matrix.

Additive setup times

Generally, you must keep in mind that setup change times are used additively if multiple setup change criteria are valid when an operation is planned. It makes no difference here whether or not they are the result of different setup change criteria (e.g. for color and tool) or if they are created within a setup change criterion (e.g. tool) when wildcards are used.

Example using different setup change criteria:

Type	From	To	Dyn. setup time
TOOL	WZ1	WZ2	3:00
TOOL	WZ2	WZ1	5:00
COLOR	BLUE	YELLOW	3:00
COLOR	GREEN	YELLOW	1:00

OP 1		OP 2	
- TOOL:	WZ1	- TOOL:	WZ2
- COLOR:	YELLOW	- COLOR:	BLUE

If OP 1 is planned after OP 2, it is given the dynamic (additional) setup time: 5:00 + 3:00 = 8:00 hours. If, on the other hand, they are planned in the opposite order (OP 2 after OP 1), then the dynamic setup time is only 3:00 hours.

Using the wildcards * and ?

You can also use wildcards in the setup change matrix:

* as a placeholder for random characters

? as a placeholder for exactly one random character.

Entering [from * to *] means that the dynamic setup time entered in the relevant row is added to any value. However, if the value of the preceding OP corresponds to that of the subsequent OP, the dynamic setup time entered in the relevant row will not be added.

Entering [from * to <value>] means that the dynamic setup time is added to any value of the preceding operation if the subsequent operation includes the value <value>. Vice versa, entering [from <value> to *] means that the dynamic setup time is added to any value of the subsequent operation, if the preceding operation includes the value <value>.

Please note to add the wildcard * as a suffix.

Please note: if the value of the preceding OP corresponds to that of the subsequent OP, a dynamic setup time will be added - in contrast to if [from * to *] is entered. If you want to ignore the dynamic setup time for identical values, you can enter an additional value with negative dynamic setup time. Please also compare the below-mentioned example "change of color".

Using the wildcard #

If you configure the entry [from # to #], then the dynamic setup time is added if the value of the preceding OP and the value of the succeeding OP are identical.

If you enter [from ##?? to ##??] in the setup change matrix, the dynamic setup time will be added for each value of the preceding operation if the third and fourth character of the subsequent OP are different. For example: **GR**EEN -> **GR**AY meets the requirement. **BR**OWN -> **BL**UE does not meet the requirement.

Generally when you use wildcards, keep in mind that these kinds of setup change entries are not given any priority when a search is run, but are instead treated as of equal value to other setup change entries so that any other valid setup change entries are integrated as well, and their dynamic setup times are also added.

Example:

OP 1 TOOL: WZA1

OP 2 TOOL: WZA1

OP 3 TOOL: WZA2

OP 4 TOOL: WZB1

OP 5 TOOL: WZBX

Setup change matrix:

Type	From	To	Dyn. setup time
TOOL	WZA*	WZB*	4:00
TOOL	WZB*	WZA1	3:00
TOOL	WZB*	WZA2	2:00
TOOL	WZB*	WZ?1	1:00

Planning (sequence):

Preceding OP	Tool	Succeeding OP	Tool	Results in dynamic setup time
OP 1	WZA1	OP 2	WZA1	0:00
OP 1	WZA1	OP 3	WZA2	0:00
OP 1	WZA1	OP 4	WZB1	4:00
OP 1	WZA1	OP 5	WZBX	4:00
OP 3	WZA2	OP 1	WZA1	0:00
OP 3	WZA2	OP 2	WZA1	0:00
OP 3	WZA2	OP 4	WZB1	4:00
OP 3	WZA2	OP 5	WZBX	4:00
OP 4	WZB1	OP 1	WZA1	3:00 + 1:00 = 4:00
OP 4	WZB1	OP 2	WZA1	3:00 + 1:00 = 4:00
OP 4	WZB1	OP 3	WZA2	2:00
OP 4	WZB1	OP 5	WZBX	0:00
OP 5	WZBX	OP 1	WZA1	3:00 + 1:00 = 4:00
OP 5	WZBX	OP 2	WZA1	3:00 + 1:00 = 4:00
OP 5	WZBX	OP 3	WZA2	2:00
OP 5	WZBX	OP 4	WZB1	1:00

Example for the color change:

OP 1 Color: White

OP 2 Color: Black

OP 3 Color: White

Setup change matrix:

Type	From	To	Dyn. setup time
COLOR	*	WHITE	10:00
COLOR	WHITE	WHITE	-10:00

Planning (sequence):

Preceding OP	Color	Succeeding OP	Color	Results in dynamic setup time

Preceding OP	Color	Succeeding OP	Color	Results in dynamic setup time
OP 1	WHITE	OP 2	BLACK	0:00
OP 1	WHITE	OP 3	WHITE	0:00 (+10:00 - 10:00)
OP 2	BLACK	OP 1	WHITE	10:00
OP 2	BLACK	OP 3	WHITE	10:00
OP 3	WHITE	OP 1	WHITE	0:00 (+10:00 - 10:00)
OP 3	WHITE	OP 2	BLACK	0:00

Including setup change times in shop floor scheduling

The static and dynamic setup times are recalculated and become visible in the OP bar if turns out during planning/replanning that the values defined in the setup change matrix match an operation value. If an OP had been planned and its static setup time was supposed to be canceled/ignored, and if this OP is now re-planned and there is no entry in the setup change matrix for this new situation, then this static setup time must be used again.

Example of a setup change matrix

Type	From	To	Addition	Ignore static setup time
Material	Mat2	Mat1	1 h	No
Color	Green	Red	2	No
Material	Mat2	Mat3	-1	No
Material	Mat2	Mat4	-3	No
Material	Mat5	Mat6	0	Yes
Material	Mat7	Mat8	-1	Yes
Color	Purple	Yellow	1.5	Yes

Planning (sequence):

The static and dynamic setup times are calculated for every second operation.

Plan OP "from - to"	Static setup time of the OP	Material	Color	Setup change matrix Y/ N = ignore static setup time 2 h etc. = dyn. setup time	Calculated stat. setup time	Setup time addition of the OP
HLS100000050	1.5	Mat2	Green			
HLS200000050	1.0	Mat1	Red	Mat1 -> Mat2; N; 1 h green -> red; N; 2 h	**1**	**3**
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HLS100000050	1.5	Mat2	Green			
HLS200000100	1	Mat3	Red	green -> red; N; 2 h Mat2 -> Mat3; N; -1 h	**1**	**1**
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HLS300000050		Mat5	Blue			
HLS400000050	1.0	Mat6	Blue	Mat5 -> Mat6; Y; 0 h	**0**	**0**
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HLS500000100		Mat7	Yellow			
HLS400000200	1.0	Mat8	Purple	Mat7 -> Mat8; Y; -1 h	**0**	**0**
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HLS400000200		Mat8	Purple			

Plan OP "from - to"	Static setup time of the OP	Material	Color	Setup change matrix Y/ N = ignore static setup time 2 h etc. = dyn. setup time	Calculated stat. setup time	Setup time addition of the OP
HLS500000100	1.0	Mat7	Yellow	purple->yellow, Y; 1.5 h	**0**	**1.5**
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HLS600000200		Mat8	Yellow			
HLS700000100	2.0	Mat7	Red	yellow->red; Y; -0.5 h	**0**	**0**
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HLS700000200		Mat8	Yellow			
HLS800000100	1.5	Mat7	Green	yellow->green; N; -0.5 h	**1.5**	**-0.5**

Consideration during lead time scheduling

The "current" static setup time is considered each time during scheduling. That is to say: after the production order is transferred from the ERP system, the system takes into account the static setup time transferred during scheduling.

When planned OPs are scheduled, the "additional setup time" is subtracted from the static setup time. Static setup time cannot be less than zero (0).

Information on user fields

Restricted selection of user fields

Altogether, you can choose from a maximum of 21 alphanumeric setup change criteria of different length and 16 numeric setup change criteria. They are provided as user fields at the operation. You can use the following user fields of the operation *(object = AGNR)* as setup change criteria:

Field ID / index	Field data type	Type	Exceptions	Number of fields
7 – 22	Numeric, time, duration	107 – 122		16
29 – 31	Text field, length 1	129 – 131		3
45 – 50	Text field, length 10	145 – 150		6
51 – 64	Text field, length 20	151 – 164	55,57,58,59	10

Field ID / index	Field data type	Type	Exceptions	Number of fields
65 – 66	Text field, length 40	165 – 166		2

Altogether, you can choose from a maximum of 6 alphanumeric setup change criteria of different length and 16 numeric setup change criteria. They are provided as user fields at the order header. You can use the following user fields of the order header *(object = AUNR)* as setup change criteria:

Field ID / index	Field data type	Type	Exceptions	Number of fields
7 – 22	Numeric, time, duration	207 – 222		16
29 – 30	Text field, length 1	229 – 230		2
45	Text field, length 10	245		1
53 – 54	Text field, length 20	253 – 254		2
60	Text field, length 20	260		1